The Palaeomagnetism of some Scottish Tertiary Dykes and their Baked Contacts

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Summary

This paper presents results of measurements of the directions of magnetization of dyke samples and the associated baked sediments collected from Ayrshire in Scotland. The results suggest that these reversely and normally magnetized rocks are due to inversions of the Earth's magnetic field rather than to any self-reversal mechanism.

1. Introduction

The reality of apparent reversals of the past Earth's magnetic field polarity can be tested by comparing the direction of magnetization of igneous rocks *and* of their associated baked rocks. The essence of the test is that if the baked rocks almost always have the same polarity as their associated igneous rocks, then there is a good case for field reversal; but if the polarities of baked and igneous rock disagree in about 50 per cent of the cases, then some physical mechanism within the rocks ought to be responsible for self-reversal, and then field reversals will not have occurred. A review of the baked rock evidence has been published by Wilson (1962). The evidence is strongly in favour of reversals of the Earth's magnetic field polarity. A particularly convincing test of ancient field polarity has been published by Everitt & Clegg (1962), who measured baked rocks at varying distances from the contact interface.

This paper presents the results of further measurements carried out by the author on samples of dyke rocks and their associated baked Permian and Old Red Sandstones collected from Ayrshire in Scotland (Lat. 56° N and Long. 5° W).

Specimens of diameter $\frac{7}{8}$ in. and length 1 in. were cored out of the samples and their directions measured with an astatic magnetometer. Each dyke specimen was then thermally demagnetized in a furnace, by heating to above its Curie point, taking measurements of directions every 50 °C. The cleaned direction of each specimen was deduced from the high temperature value on the vector heating diagrams. As a test of stability (Brunhes & David 1903), the directions of associated baked Permian sandstones were also measured after cleaning them by heating to about 300 °C and cooling in zero field, since they were too weakly magnetized to measure while hot. Table 1 shows the result of these measurements. Figure 1 is the stereographic projection of directions of single dyke samples before and after cleaning, and Figure 2 that of directions of cleaned dykes and mean directions of

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Table 1

Dyke directions (single specimens)

Specimen	Location of dyke	Before thermal demagnetization		After thermal demagnetization		Width of dyke	Type of Sandstone baked by dyke	Distance of baked sample from contact	Number of baked specimens measured after cleaning	Mean direction of baked specimens after cleaning	
		Declination east of true north	Inclination	Declination east of true north	Inclination					Declination East of true north	Inclination
R A I	Dykes	5°	77° down	346°	74° down	3 ft	Permian.	4 in.	6	282°	76° down
R A V	Crossing	132°	19° up	148°	31° up	3 ft 3 in.	Permian.	16 ft	3	146°	0°
R A VI	River	224°	62° up	214°	60° up	3 ft	Permian.	–	3	204°	51° up
R A VII	Ayr.	188°	11° up	185°	16° up	3 ft	Permian.	6 in.	3	174°	4° up
D I	Dipple on coast of Ayr.	213°	40° up	188°	62° up	9 ft	Old Red Sandstone.	2 ft	2	152°·5	58°·5 up
CU II	Culzean Bay.	123°	48° up	153°	62° up	37 in.	Old Red Sandstone Bleached grey at contact.	1 in.	2	147°	55° up
S III	Seamills	9°	15° down	301°	68° up	7 ft	Old Red Sandstone baked first by sill and then by dyke.	Right on contact.	3	234°	58°·5 up
SK 1	Skelmorlie	203°	65° up	209°	66° up	14 ft	Old Red Sandstone greatly metamorphosed.	9 ft	3	186°	49° up
IK I	Inverkip	240°	58°	233°	67° up	45 ft	Red Sandstone.	20 ft	2	223°	67°·5 up

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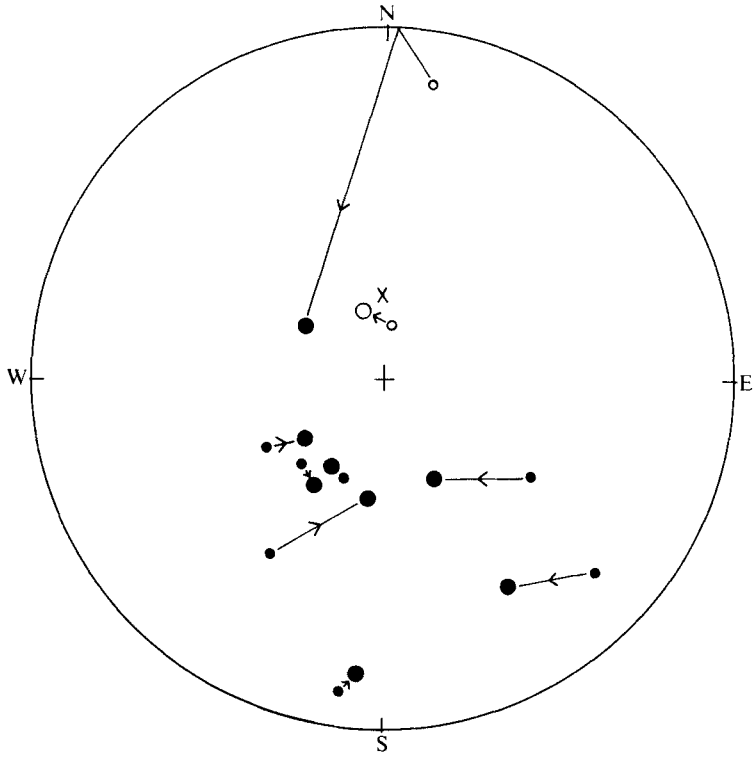


FIG. 1.—Directions of magnetization of dykes before and after thermal demagnetization.

● Dips up
○ Dips down
Solid line links the uncleaned specimen (small symbol) with the cleaned one (large symbol)
× Present dipole field

associated baked rocks. Mean directions were found by averaging the directions cosines, as suggested by Fisher (1953).

2. Results and discussion

Of the nine dyke specimens, one from each dyke, one showed normal and eight showed reversed directions after cleaning. The mean directions of the baked rocks agree fairly well with those of the corresponding dyke rocks (Figure 2), although one error made was to collect baked specimens not extremely close to the dykes, so they were not so highly baked as could be desired.

RA V and RA VII have shallow dips and may indeed be of Permian age. The rest of the dykes are judged to be of Tertiary age because of their directions of magnetization and their NNW bearings in the field. However, the baked rocks always have the same sense of magnetization as the dyke rocks, regardless of the ages of the dykes.

Table 2 gives the mean directions of British Tertiary rocks measured by Bruckshaw & Vincenz, and Blundell, Khan and Wilson. The mean axis of the

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Table 2

Author	Location and type of rock	Direction of Magnetism: Declination east of true north	Inclination
Bruckshaw & Vincenz (1954)	Mull lavas	166°	73° up
Blundell (1957)	Lundy and Lee dykes	197°	59° up
Khan (1960)	Skye–Lavas and intrusives	186°·1	59°·7 up
	Ardnamurchan–Gabbro	184°·1	54° up
	Rum–Layered Gabbro	181°·0	55°·1 up
Wilson (1961) and Hospers & Charlesworth (1954) (Combined result)	Northern Ireland Basalt lavas, dykes, sills and non-igneous rocks	188°	62°·8 up
Present result of dyke measurements	Dykes from Ayrshire, Scotland	189°	62°·7 up

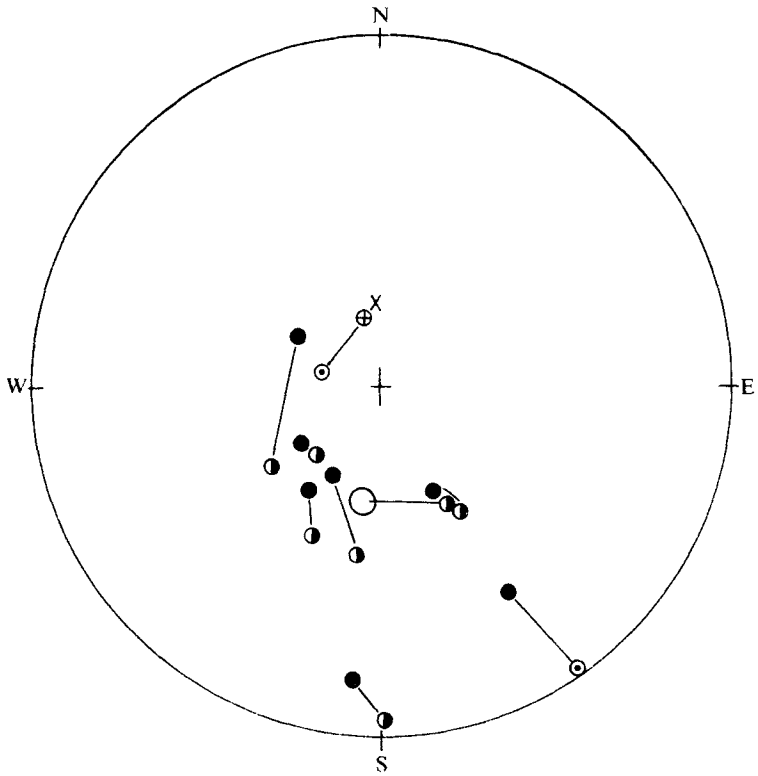


FIG. 2.—Directions of dykes and associated baked rocks.
○ Mean dyke direction ● Dyke with dip up
⊕ Dyke with dip down ◑ Baked rock with dip up
⊙ Baked rock with dip down
Solid line links the dyke with the associated baked specimen
× Present dipole field

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nine dykes is directed 189° (E.T.N.), 62·7° up. This compares well with the results of the previous authors given in Table 2.

The agreement of sense of dykes and associated baked samples supports the idea of reversals of the Earth's magnetic field, and suggests that the reversed dykes are not due to some self-reversing mechanism.

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